

$$u_L = \frac{\int d\Phi |M_L|^2}{\int d\Phi |M|^2} \rightarrow \gamma^0$$

$$\sigma = \left(\int d\Phi_B (|M_B|^2 + 2\text{Re}\{M_B M_V\}) + \int d\Phi_3 |M_A|^2 \right) - D$$

$M_{B \text{ like}} + D \rightarrow \gamma^0$ maybe?

$$\text{reweight } r_L^{BV} = \frac{M_{B \text{ like}}^L}{M_B^{\text{line}}}$$

$$\begin{cases} B \rightarrow \gamma_L^{BV} \\ B+j \rightarrow \gamma_L^R \\ B \\ B \end{cases}$$

factorisation of IR in QCD should work.

Event generated almost always with Rad. because of resummation.

$$\tilde{B}_{\text{PWG}} = \left[B + V_{\text{subtr}} + \int d\Phi_{\text{rad}} R_{\text{subtr}} \right]_h$$

Alternatively:

- compute $\tilde{B}_{\text{NLO}}$ for UNP

$$e^- \sim e^- \frac{\int d\Phi_{\text{rad}} R}{\sqrt{d\Phi_B B}}$$

Sud.

LHEVs have rad most of times

$$q\bar{q}' \rightarrow W^+(e^+ \nu) Z(\mu^+ \mu^-) g$$

$$gq/g\bar{q}'$$

$$v_{LL}^{(n)} = \frac{|R_{n+1}^{(LL)}|^2}{|R_{n+1}^{\text{unp}}|^2}$$

$\rightarrow \frac{\infty}{\infty}, \frac{0}{0}$ risk

$$q\bar{q}' \rightarrow \nu(e\nu) \bar{\nu}(e'\bar{\nu}) \rightarrow v_{LL}^{(n)} = \frac{|B_n^{(LL)}|^2}{|B_n^{\text{unp}}|^2}$$

$\hookrightarrow$ if work 😊

focus ZZ

↳ events:

$\left\{ \begin{array}{l} \nu\nu + j \rightarrow r_{n+1}^{LL} \\ \nu\nu \rightarrow r_n^{LL} \\ \nu\nu + j \\ \nu\nu + j \\ \vdots \\ \vdots \end{array} \right.$

input: 4 leptons, r^{LL}

↳ training before PS.

↳ warm start after PS.